Total Marks: 70

Time: 90 minutes

Examiner: Miss B Hannweg Moderator: Mrs D Webb

Instructions:

1. This question paper consists of 8 pages. Please ensure you have all pages.
2. All questions are to be answered on the question paper in the spaces provided. No extra paper is to be used.
3. Ensure your answer paper has your name on it.
4. It is in your own interest to write neatly and legibly.
5. **You may not use typex.**

NAME:

BACKGRAOUND INFORMATION

The chemicals used in labs, have different natures, which can be acidic, basic or neutral. This characteristic of the chemical depends on the release of H^+or OH^- ions in its aqueous solution. Chemicals which releases H^+ ions are acidic whereas which releases OH^- are basic in nature.

1. What is pH?

pH: It is the power of H^+ ions in a solution. The term pH refers to the "potential of hydrogen ion." The pH scale is logarithmic, meaning that an increase or decrease of an integer value changes the concentration by a ten fold. For example, a pH of 3 is ten times more acidic than a pH of 4. Likewise, a pH of 3 is one hundred times more acidic than a pH of 5. Similarly a pH of 11 is ten times more basic than a pH of 10.

2. How is pH measured?

pH scale: pH is normally measured in a range of 0-14.

- **pH < 7:** The solutions having pH value below 7 are acidic in nature. These solutions turn blue litmus red.
- **pH = 7:** The solutions having pH equal to 7 are neutral in nature. These solutions do not show any colour change of litmus.
- **pH > 7:** The solutions having pH more than 7 are basic in nature. These solutions turn red litmus blue.

3. Measuring pH**:** The pH of a solution can be determined using an instrument like pH meter or various indicators which changes their colour accordingly depending upon the nature of test solution. Indicators are a type of substance that can be used to identify the nature of chemicals due to its colour change. The indicator changes its colour when the pH of the solution slightly changes.

4. Types of indicators: Litmus paper - Litmus is a dye extracted from various species of lichen. Lichens are a symbiotic partnership of two organisms, a fungus and an algae that changes colour in acid and base. Red litmus turns blue in basic and blue litmus turns red in acidic medium. The colour change of these indicators in different medium is given below:

Indicator	Acidic Medium	Basic Medium	Neutral Medium
Blue litmus	Red	Blue	Blue
Red litmus	Red	Blue	Red

MATERIALS

(a) 4 test tubes each with a prepared solution in it Labeled A-D
(b) A test tube rack
(c) A white tile
(d) A pair of forceps
(e) A paper towel
(f) Four pieces of red litmus paper and four pieces of blue litmus paper

PROCEDURE

1. Ensure you have all the correct materials on your desk.
2. Place the white tile behind the test tubes and record the colour of the solutions in each test tube in *Table A* (provided below).
3. Using the forceps, take a piece of blue litmus paper and dip it into test tube A.
4. Take the litmus paper out and place it on the paper towel.
5. Record your observation of the colour of the litmus paper and record this in *Table A.*
6. Repeat steps 3-5 with test tubes B,C and D.
7. Once you have finished with your blue litmus paper, conduct steps 3-6 with the red litmus paper. Record your observations in *Table A.*

QUESTIONS

Answer all questions in the spaces provided.

1. *Table A observations* (18)

   Heading:

Test tube	**Initial colour**	**Blue litmus paper final colour**	**Red litmus paper final colour**

2. Construct a hypothesis for this experiment. (3)

3. State your aim for this experiment. (2)

4. Name the:

   4.1. independent variable: (2)

   4.2. dependant variable: (2)

5. What are three variables which should be controlled in this experiment? (3)

6. Identify two errors that can occur when conducting this experiment. (2)

7. In your own words, describe what pH is and why it is important to measure the pH of certain substances. (3)

8. Litmus paper is made from lichen which is a symbiotic partnership between algae and fungi.

   8.1. Explain what a symbiotic relationship is. (2)

   8.2. Give an example of a symbiotic relationship found in nature. (2)

   8.3. What is it called when one organism benefits from living off another organism whilst harming it? (1)

8.4. Name one drawback each organism would face in this symbiotic relationship from question 7.3. (2)

9. The solutions provided in the test tubes for your experiment were as follows:
    - A dilute lemon juice solution
    - A diltute bicarbonate of soda solution
    - Distilled water
    - A dilute HCL solution

    Using your observations, comple the table below and indicate which test tube contained which solution: (6)

Heading:

Test tube letter	**Solution**

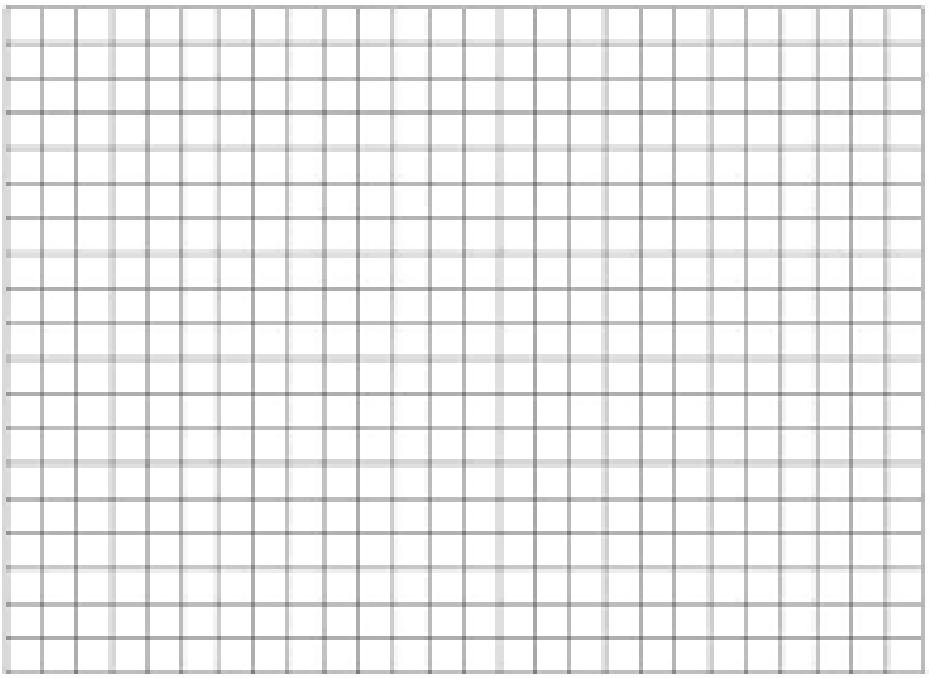
10. Explain why you assigned each solution to your chosen test tube. (4)

11. The pH of the four solutions is as follows:

- 2.6;
- 3.4;
- 7.0;
- 9.5

Using your results from the table in question 8, construct a graph on the graph paper provided below of the pH values found in each test tube. (8)

Heading:



12. Think about how basic and acidic substances contribute towards our survival everyday.

12.1. Using an example of an acidic substance found within our bodies, How would these be important to us? (3)

12.2. Using an example of an external acidic substance (i.e. found in the environment), how would this be important to our everyday lives? (3)

12.3. Using your knowledge of nutrition, what might some of the implications be of having a daily diet that is too high in acidic or basic substances? (4)